

ANDREWS K. BOAHEN

East Lansing, MI, USA

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EDUCATION

Michigan State University, East Lansing, MI Aug 2022 – May 2027 (Expected)

PH.D. STATISTICS (*Candidacy achieved*)

M.S. STATISTICS (*May 2026*)

Dissertation: *Nonlinear Computer Model Calibration: Active Learning, Inference and Theoretical Properties*

Advisor: Dr. Chih-Li Sung

African Institute for Mathematical Sciences (AIMS-Ghana), Accra, Ghana Oct 2020 – Jul 2021

M.SC. MATHEMATICAL SCIENCES (*with distinction*)

Advisor: Prof. Kwabena Doku-Amponsah

University of Ghana, Accra, Ghana Sep 2015 – May 2019

B.SC. ACTUARIAL SCIENCE (*first-class honours*)

Advisor: Prof. Kwabena Doku-Amponsah

RESEARCH INTERESTS

Computer experiments, computer model calibration, uncertainty quantification, Gaussian and deep Gaussian processes, surrogate modeling, active/sequential learning, Bayesian optimization, and statistical learning for scientific applications.

PUBLICATIONS & MANUSCRIPTS

1. **Boahen, A.K.** and Sung, C.-L. Convergence Results for Nonlinear Calibration under Squared Exponential and Matérn Kernels. (*Manuscript in preparation*)
2. **Boahen, A.K.**, Heo, J. and Sung, C.-L. Active Learning for Nonlinear Computer Model Calibration. (*Manuscript in preparation*)
3. **Boahen, A.K.**, Katsekor, T. and Doku-Amponsah, K. (2025). Large deviations for spatial telecommunication systems: The boolean model. *Journal of Information and Optimization Sciences*, 1–16. doi/10.47974/JIOS-1338.
4. **Boahen, A.K.**, Davis IV, W. L. (2024). In Situ Machine Learning for Intelligent Data Capture and Event Detection. In *Computer Science Research Institute Summer Proceedings 2024*. M. B. P. Adams, T. A. Casey, and B. W. Reuter, eds. Technical Report SAND2024-16688O. Sandia National Laboratories, pp. 288–297.

SELECTED TALKS

1. **2026 ISNET-12 Conference** (Accepted talk, October 2026). *Active Learning for Nonlinear Computer Model Calibration*. Lawrence Berkeley National Laboratory, CA.
2. **IMSI Summer Internship Workshop** (August 2026). *Network-Based Statistics for Rodent Connectomics*. Online.
3. **SIAM Conference on Uncertainty Quantification (UQ26)** (March 2026). *Active Learning for Nonlinear Computer Model Calibration*. Minneapolis, MN.
4. **15th International Conference on Monte Carlo Methods and Applications** (July 2025). *Active Learning for Nonlinear Calibration*. Chicago, IL.
5. **IMS/ASA Spring Research Conference** (May 2025). *Active Learning for Nonlinear Calibration*. New York, NY.

6. **CSRI Lightning Talks** (July 2024). *In Situ Machine Learning (ISML) for Intelligent Data Capture and Event Detection*. Sandia National Laboratories, NM.

RESEARCH EXPERIENCE

Michigan State University, Department of Statistics and Probability

May 2024 – Present

Research Assistant

- Developed the Non-Additive Calibration (NAC) framework for integrating computer simulation outputs with physical observations, extending computer model calibration beyond conventional additive discrepancy formulations.
- Developed active-learning and sequential-design strategies for nonlinear computer model calibration, including derivative-free and stochastic optimization methods for determining where additional simulation or physical data are most informative.
- Established theoretical properties of nonlinear calibration estimators, including consistency and convergence rates under Gaussian and deep Gaussian process-type surrogate models.
- Developed computational and simulation frameworks for evaluating calibration, prediction, and uncertainty quantification across complex computer experiments.
- Developing an R software package implementing the proposed nonlinear calibration, inference, prediction, and active-learning methodology.

Beckman Institute for Advanced Science and Technology, Gritton Lab

May 2026 – August 2026

Graduate Intern, IMSI Summer Internship Program

- Developed statistical learning methodology for detecting group-level differences in high-dimensional brain-connectivity data, addressing severe power limitations of standard multiple-testing procedures across thousands of network edges.
- Developed an adaptive logistic-lasso framework that integrates functional and structural brain-connectivity information, using structural connectivity as adaptive prior information for high-dimensional group comparison and prediction.
- Built an end-to-end Python pipeline using NumPy, SciPy, and nilearn for connectivity estimation and large-scale permutation inference, contributing to an ongoing methodological manuscript.

Sandia National Laboratories, Scientific Machine Learning Department

June 2024 – August 2024

Scientific Machine Learning Intern

- Developed manifold-learning and dimension-reduction methods for real-time detection and characterization of anomalous events in high-dimensional nonlinear scientific simulations, extending Sandia's In Situ Machine Learning (ISML) framework for complex dynamical systems.
- Delivered a lightning talk on ISML algorithms and manifold-learning signatures for real-time event detection to ~70 interns and researchers at Sandia's CSRI Summer Symposium.
- Published technical report, *In Situ Machine Learning (ISML) for Intelligent Data Capture and Event Detection*, in CSRI Summer Proceedings 2024 (SAND2024-166880).

African Institute for Mathematical Sciences, Probability Group

January 2022 – August 2022

Research Master Student

- Improved extreme value index estimation in Pareto-type distributions by developing a generalized Pickands estimator that reduced RMSE relative to Hill and bias-corrected Hill competitors across multiple regression frameworks.
- Rigorously characterized the generalized Pickands estimator through theoretical derivations and simulation studies, establishing consistency, unbiasedness, asymptotic normality, and convergence rates under Pareto-type tail assumptions.

HONORS & AWARDS

- (2026) Selected as a **finalist for the IMSI Summer Graduate Student Internship** and funded to attend the Spring Coding and Data Science Bootcamp.
Awarded the Michigan State University **Graduate School Travel Fellowship**.
- (2025) Recipient of the **JMP-P&G Student-Early Career Travel Award**.
Elected as a **full member of Sigma Xi**, The Scientific Research Honor Society.
- (2024) Received a **Sustainable Research Pathways (SRP) Fellowship**.
- (2022) Awarded the **Research Master Student Scholarship** as the best graduating research master student at AIMS-Ghana.
- (2021) **Valedictorian, summa cum laude**, AIMS-Ghana graduating class.
Awarded the **F.K.A. Allotey Meritorious Award**.
- (2020) Awarded the **African Institute for Mathematical Sciences Scholarship**.

TEACHING EXPERIENCE

Michigan State University <i>Instructor / Graduate Teaching Assistant / Grader</i>	Aug 2022 – Present
• Taught and supported undergraduate and graduate courses in statistical methods, Bayesian statistics, statistical inference, probability theory, and multivariate data analysis. • Instructor of record for STT 200 (Statistical Methods) during Summer 2023.	
University of Ghana <i>Teaching Assistant</i>	Aug 2019 – Aug 2020
• Supported courses in multivariate methods, biostatistics, probability, survival analysis, and stochastic processes.	

SERVICE & PROFESSIONAL AFFILIATIONS

Professional Memberships	IMSI, ASA, SIAM; Sigma Xi (Full Member)
General Secretary	African Graduate Student Association (2026–2027)
Elected Student Representative	MSU Statistics Colloquium Committee (2025–2026)
Graduate Student Instructional Leader	MSU Department of Statistics & Probability (2025–2026)
Graduate Student Mentor	MSU Department of Statistics & Probability (2023–2025)
Judge/Evaluator	28th Annual UURAF & MidSURE

SKILLS

Statistical Learning & UQ	Gaussian and deep Gaussian processes, uncertainty quantification, Bayesian and variational inference, high-dimensional statistical inference.
Optimization & Design	Active/sequential learning, Bayesian optimization, computer model calibration, stochastic optimization, MCMC/SGD methods, optimal experimental design.
Scientific Computing Software	R, Python (PyTorch, NumPy, SciPy, scikit-learn, nilearn), TensorFlow, HPC. Git/GitHub, statistical software development, simulation and uncertainty-quantification pipelines.
Languages	English (Fluent), French (Fluent).